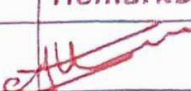
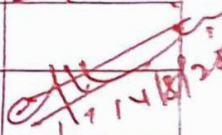
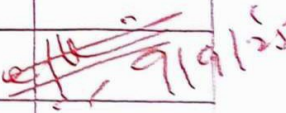


I N D E X

NAME: Kuberan STD.: A1 SEC.: A ROLL NO.: _____ SUB: Deep learning

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
1	24/7	Exploring Deep learning Platform		
2	31/07	Implement a classifier using open source dataset		
3	14/08	study of the classifiers with respect to statistical parameters		
4	14/08	Build a simple feed forward neural network to register uc		
5	22/08	study of Activation function and its role		
6	09/09	Gradient descent & backpropagation in deep neural networks		
7	16/09	Build a cnn model classify cat and dog image		
8	9/10	experiment of LSTM		
9	29/10	Build a Recurrent Neural Networks		
10	17/10	Perform Compression on mnist dataset using Autoencoder		
11	17/10	Experiments using variational Autoencoder (VAE)		
12	26/10	Implement a Deep convolution GAN (DCGAN) to color image		
		Understand the Architecture of trained Model		

17/p/25

11. Experiments Using Variational Autoencoder (VAE)

Aim

To implement a Variational Autoencoder (VAE) and study its generate ability to reconstruct and generate new data samples using the MNIST dataset.

Objectives

1. To understand the concept and working of a Variational Autoencoder (VAE)
2. To generate new data points by sampling from a latent distribution
3. To visualize reconstructed and newly generated images
4. To compare the performance of VAE with a basic Autoencoder.

Pseudocode

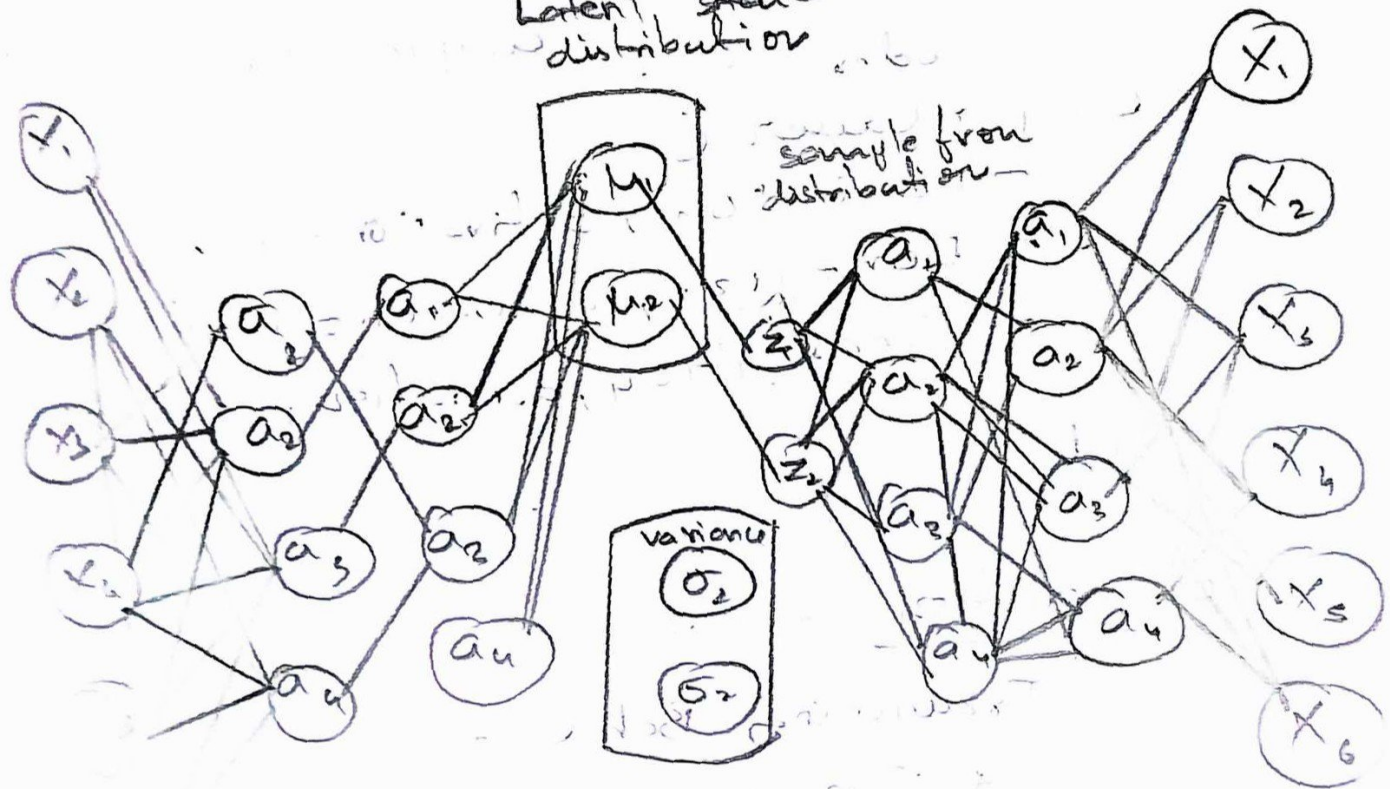
Start

1. Import libraries (TensorFlow/Keras, numpy, matplotlib)
2. Load MNIST dataset and normalize images (0-1)
3. Define Encoder
 - Input layer (784)
 - Dense (256, activation = 'relu')

Autoencoder (VAE)

Latent state distribution

sample from distribution



ENCODER

DECODER →

Dense (128, activation = 'relu')
- output : mean (μ) and log
variance (σ^2)

4. Sample latent vector z using
reparameterization:

$$z = \mu + \sigma * \epsilon, \text{ Where } \epsilon \sim N(0, 1)$$

5. Compile and train model us
MNIST dataset

End

observation

1. Training Dynamics:

* The VAE learned to balance
between accurate reconstruction and
maintaining a continuous latent space

2. Latent Space Representation.

* Unlike normal autoencoder
the VAE's latent space followed a
Gaussian distribution

3. Reconstruction Quality

* The model effectively learned
the key structural pattern of digits

4. Generative Ability:

* New samples generated
by sampling random vectors

Result

successfully Implemented a
Variational Autoencoder (VAE) on the MNIST
dataset

output:

Epoch 1, Average loss: 164.3244

Epoch 2, Average loss: 121.2762

Epoch 3, Average loss: 114.4349

Epoch 4, Average loss: 111.4662

Epoch 5, Average loss: 109.7881

Epoch 6, Average loss: 108.594

Epoch 7, Average loss: 107.7130

Epoch 8, Average loss: 107.1150

Epoch 9, Average loss: 106.5510

Epoch 10, Average loss: 106.1544